

ASSIGNMENT 1, FINANCIAL ECONOMETRICS

This assignment will help you apply the concepts discussed in class. Submit it on Canvas. Follow the instructions. Some of you may want to work individually. However, you can work in groups of up to three students. Read all assignment instructions, including the notes at the end of this document.

1. Data

The analysis will use the time series listed below. You will not need data files because the code downloads the data directly from a public source. You will be working with two ETF time series.

Code

The assignment will require R code. The assignment materials include the required R programs, which you should run as follows:

a. *Decomposition*

- The first program runs the R decomposition code, which shows the various components, as discussed in class, on each of the two time series. The code separates the trend and cyclical components in the original R output, as learned in class, and generates a four-section chart showing the trend, cyclical, seasonal, and noise components.

b. *Stationarity*

- The second R program runs the ADF test on each time series to determine whether you can or cannot reject the null hypothesis that the applicable series is non-stationary. The stationarity test outputs appear in the lower-left window of the RStudio interface.

c. *White Noise*

- The third R program runs the Ljung-Box test on the applicable time series to determine whether to reject the null hypothesis that the series is white noise. Together, the ADF and Ljung-Box tests indicate how far we can consider these series stationary.

d. *ACF Output*

- This program produces the Autocorrelation Function (ACF) for the return series.

You will record the results of the four R programs on a “report template,” which is included with the assignment materials. The report template will include spaces to paste the outputs and text boxes to discuss the meaning of the results.

e. Discussion Questions

Based on the results you obtained from running the R code, you will address a set of questions at the end of the report template.

Important: All answers to the discussion questions must be your own and in your own words. Record the answers to the questions in your report. If you work as a team, prepare and submit only one team report, with the team members' names on the cover. Using AI tools to do any portion (or all) of the work for you violates the code of conduct. In the case of these assignments, if you use AI tools as a study tool, for example, to confirm definitions or concepts, disclose it at the end in a reference section. You can also reference econometric papers or books. If you use these references, cite them using a common reference style, such as APA or Chicago.

f. Completing the Report

As noted above, the outputs, analysis, and responses to the discussion questions will be summarized in the attached report template. You will submit this report as a **PDF document**. Save it as a PDF file before submitting to avoid issues when opening it in Canvas. Do not submit code or data.

Other Notes

- **Assume you are preparing this report for presentation to a group of colleagues interested in the topic. Use the provided template and maintain a professional report style.**
- **Be clear and organized when discussing the results or answering the questions. Clarity and organization will factor into the evaluation.**
- **As noted earlier, when answering the discussion questions, you may cite papers, books, or other documented sources to support your explanations or arguments (if you consider it necessary). Copying and pasting materials from a search engine or using AI tools do the work for you (e.g., discussions or opinions) is prohibited and violates the JHU student conduct code. You can use AI tools to consult definitions and concepts, with proper referencing. The instructor may use tools to identify content obtained through browser or artificial intelligence sources.**
- **Submit the report in PDF format. Canvas works better with PDF files. You only need to submit your PDF file. Do not submit CSV or R files.**
- **You can work in groups of up to three students. Submit only one report per group. If you work in groups, ensure all student names are on the report cover.**
- **Whether you work individually or as a team, do not share outputs or answers to the discussion questions with other students. If you work individually, keep the solutions in your individual report. If you work as a team, prepare the team's solutions and record them in your team report. Again, submit only one report for your team, with all team members' names on the PDF report cover.**
- **Start early to have sufficient time to assemble the materials, including the answers to the discussion questions. Putting the report together will take time.**